

UNIVERSIDAD TECNOLÓGICA DE PANAMÁ
Facultad de Sistemas Computacionales
Asignatura: Programación II
Examen Parcial1

Profesor: Napoleón Ibarra

Estudiante: Magdalena González

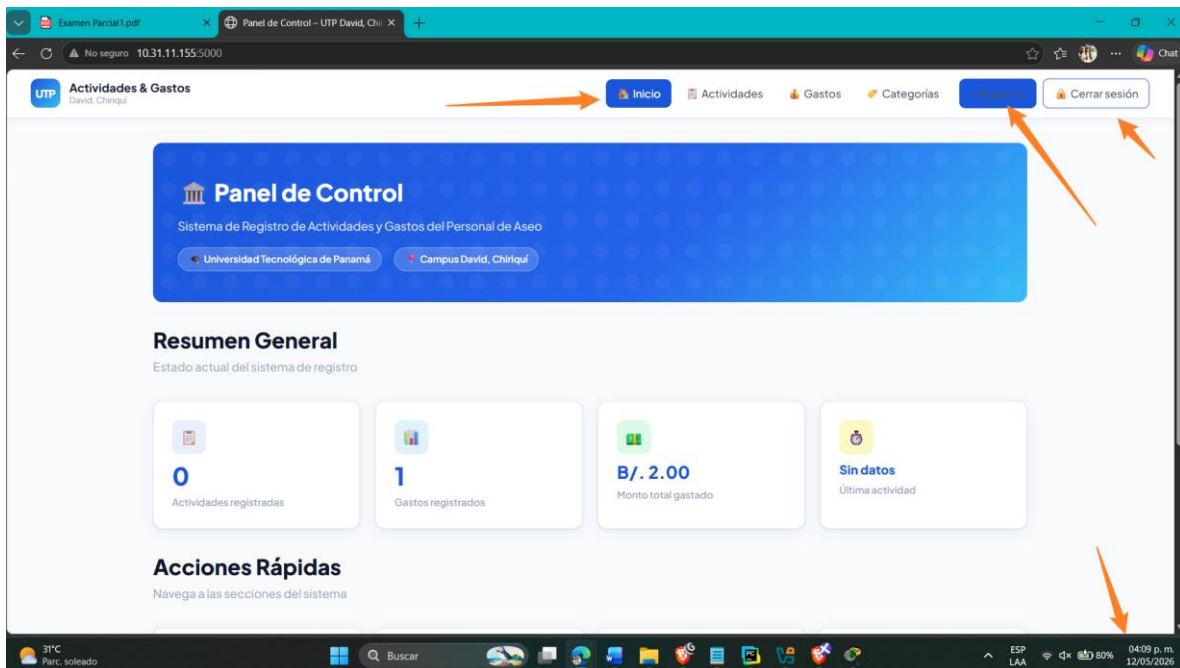
Cédula: 4-819-1590

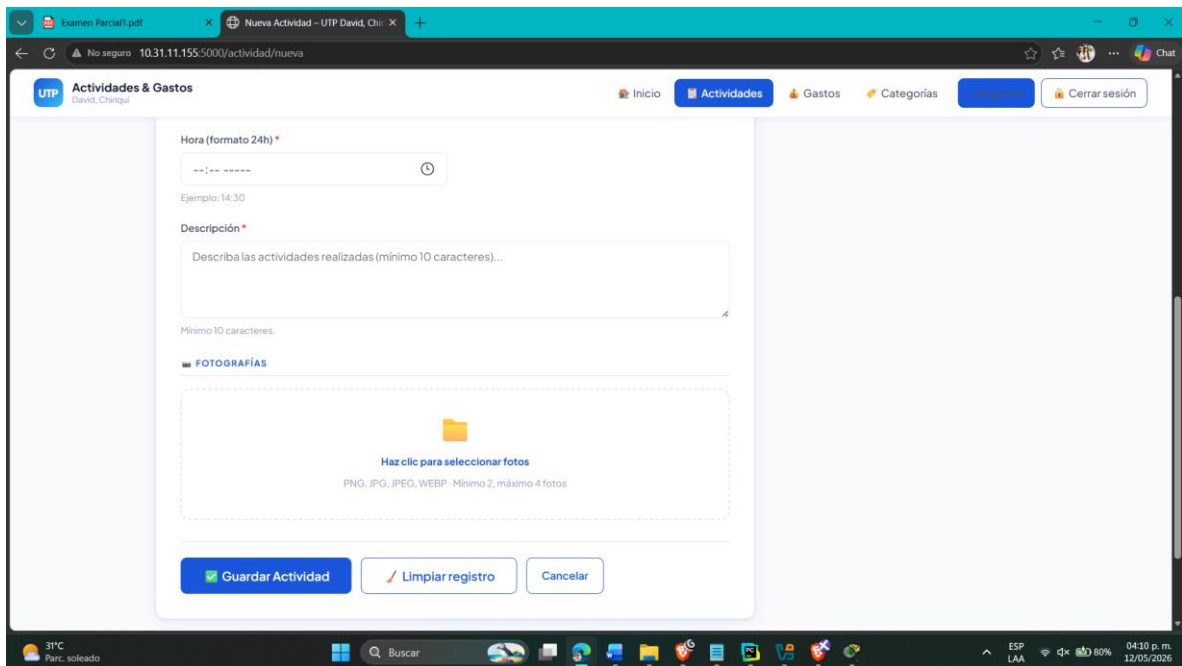
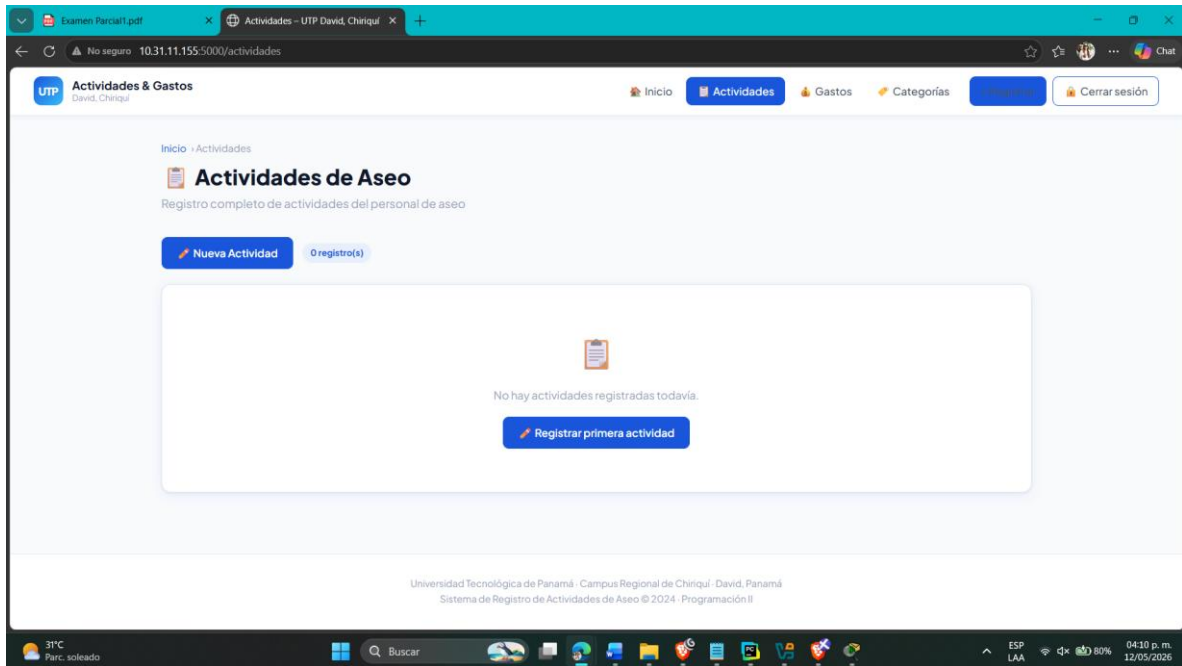
Fecha de Inicio: 12/05/2026 – Hora: 2:30 PM

Fecha de Entrega: 12/05/2026 – Hora: 4:55 PM

Fecha de Entrega: 12/05/2026 – Hora: 5:00 PM

I Parte. Caso de Estudio. Valor 30 puntos





Examen Parcial1.pdf

Gastos - UTP David, Chiriquí

No seguro

10.31.11.155:5000/gastos

Chat

UTP

Actividades & Gastos

David, Chiriquí

Inicio

Actividades

Gastos

Categorías

Logout

Cerrar sesión

Inicio > Gastos

Gastos Operativos

Registro de gastos del personal. Filtro por categoría y totales automáticos.

Nuevo Gasto

Todas las categorías

1 registro(s)

Total General

B/. 2.00

#	DESCRIPCIÓN	CATEGORÍA	MONTO	FECHA	OBSERVACIÓN	REGISTRADO	ACCIONES
1	escobas	Comida	B/. 2.00	2026-05-28	ASCVBGHML...	2026-05-12 15:30:36	
Total General			B/. 2.00				

Universidad Tecnológica de Panamá - Campus Regional de Chiriquí - David, Panamá
Sistema de Registro de Actividades de Aseo © 2024 - Programación II

31°C

Parc. soleado

Buscar

ESP

LAA

80%

04:10 p. m.

12/05/2026

Examen Parcial1.pdf

Categorías de Gasto - UTP David, Chiriquí

No seguro

10.31.11.155:5000/categorias

Chat

UTP

Actividades & Gastos

David, Chiriquí

Inicio

Actividades

Gastos

Categorías

Logout

Cerrar sesión

Inicio > Categorías

Categorías de Gasto

Administra las categorías disponibles para registrar gastos.

Nueva Categoría

6 categoría(s)

#	NOMBRE	GASTOS ASOCIADOS	FECHA DE REGISTRO	ACCIONES
1	Comida	1 gasto(s)	2026-05-12 16:06:44	No eliminable
4	Limpieza	Sin gastos	2026-05-12 16:06:44	Eliminar
3	Materiales	Sin gastos	2026-05-12 16:06:44	Eliminar
6	Otros	Sin gastos	2026-05-12 16:06:44	Eliminar
5	Transporte	Sin gastos	2026-05-12 16:06:44	Eliminar
2	Viáticos	Sin gastos	2026-05-12 16:06:44	Eliminar

31°C

Parc. soleado

Buscar

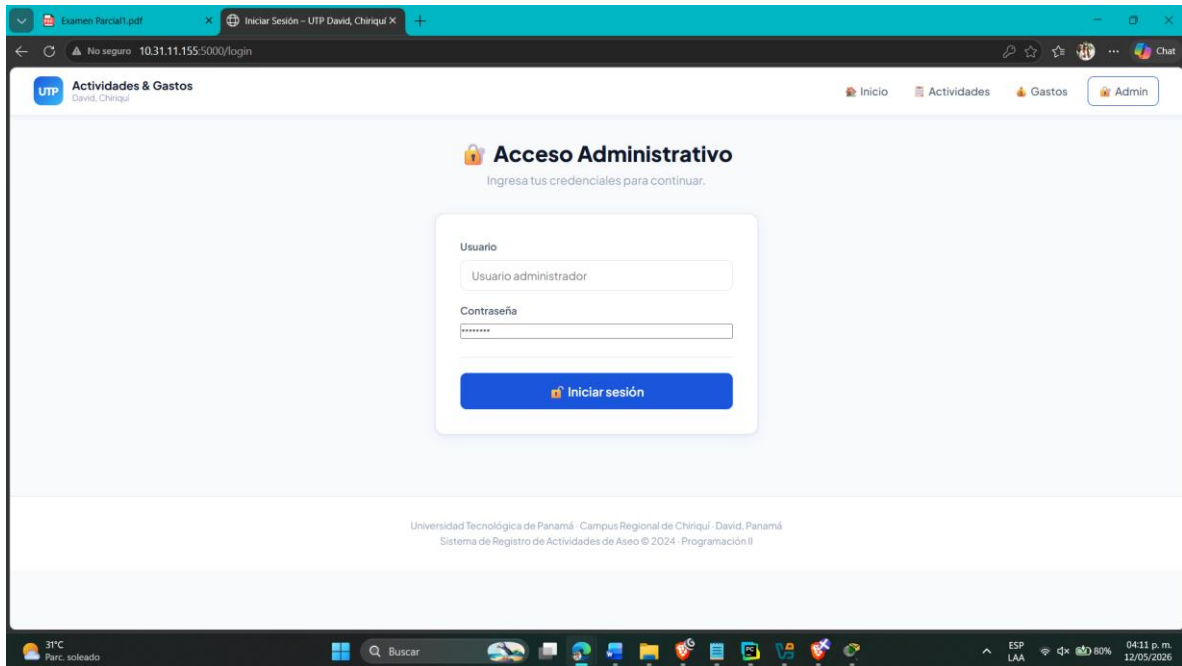
ESP

LAA

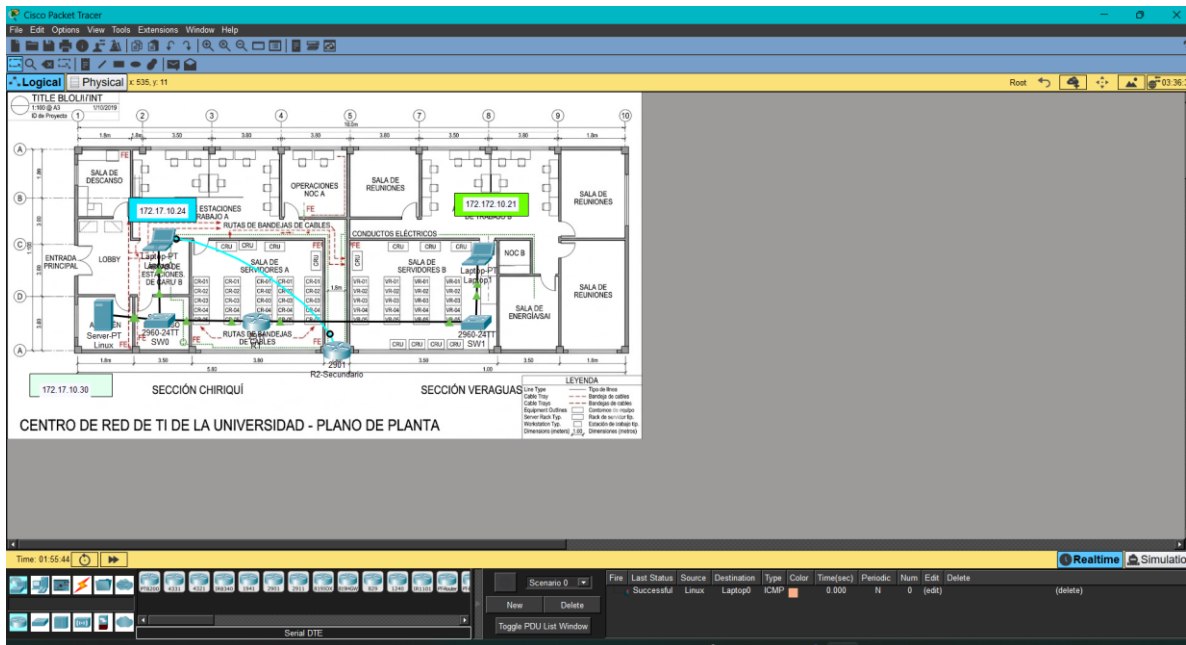
80%

04:10 p. m.

12/05/2026



II PARTE. Simulación de Red LAN. Valor 20 puntos

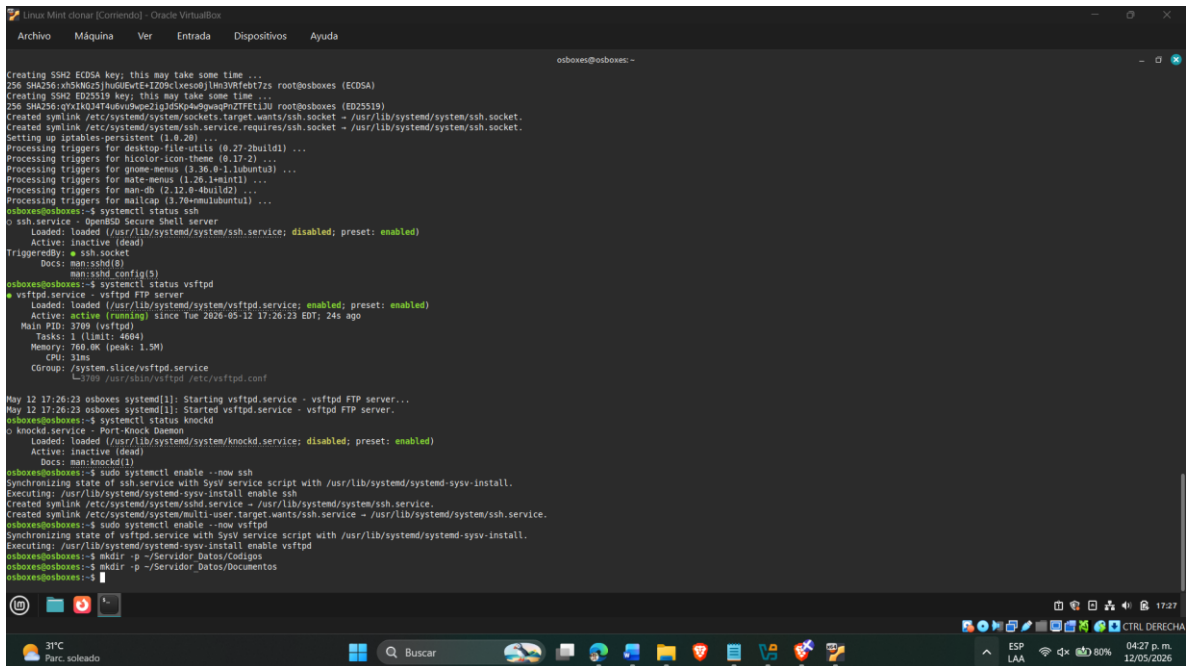
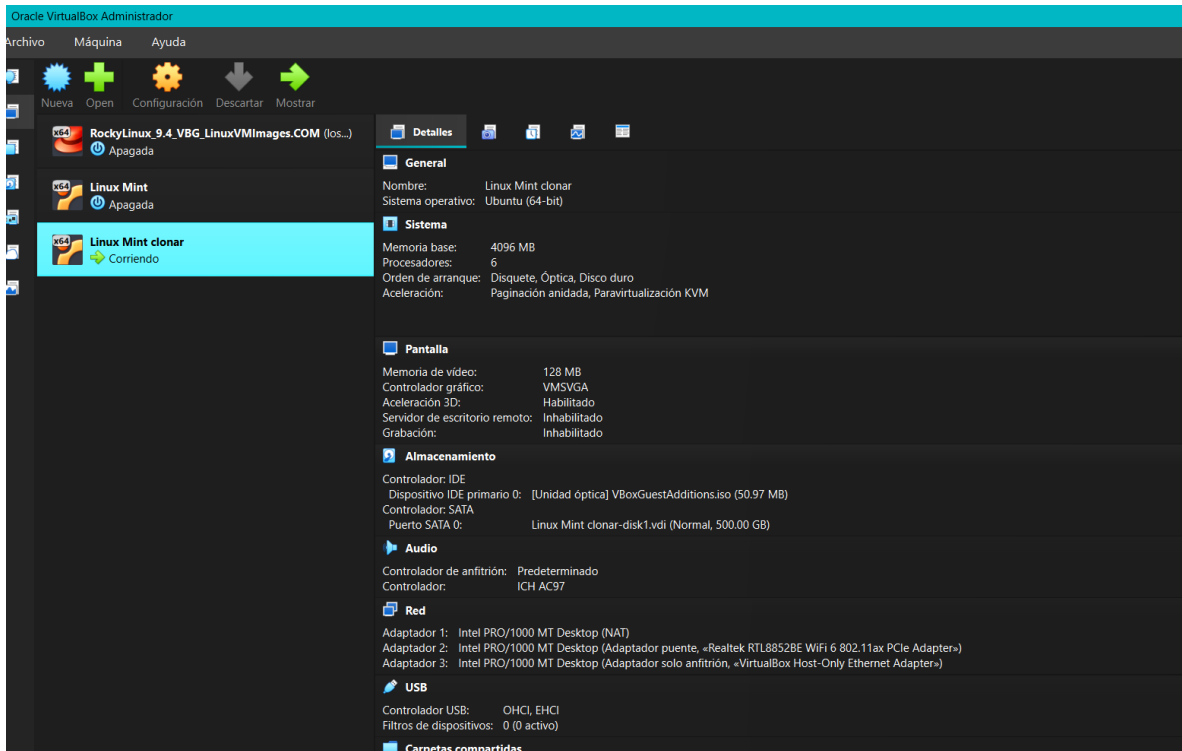




SECCIÓN 2.pkt

NOMBRE DISPOSITIVO	DEL	IP ADDRESS	MÁSCARA SUBRED	DE
R1		172.17.10.1 / 172.172.10.1	255.255.255.0	
SW1		172.17.10.2	255.255.255.0	
SW2		172.172.10.2	255.255.255.0	
L1		172.17.10.24	255.255.255.0	
L2		172.172.10.21	255.255.255.0	
SERVER1		172.17.10.30	255.255.255.0	

III PARTE. Caso de Estudio. Valor 20 puntos




```
ComputerName      : 192.168.56.102
RemoteAddress     : 192.168.56.102
RemotePort        : 22
InterfaceAlias    : Ethernet 2
SourceAddress     : 192.168.56.1
PingSucceeded     : True
PingReplyDetails  (RTT) : 1 ms
TcpTestSucceeded  : False
```

```
P:\C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> Test-NetConnection 192.168.56.102 -Port 22
ADVERTENCIA: TCP connect to (192.168.56.102 : 22) failed
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> C:\Users\Sari8\OneDrive\Documents\ParcialProgramacionMG\ExamenParcial1_Flask
```

```
PS C:\WINDOWS\system32> Test-NetConnection 192.168.56.102 -Port 7000
ComputerName : 192.168.56.102:102 : 7000 failed
RemoteAddress : 192.168.56.102
RemotePort : 7000
InterfaceAlias : Ethernet 2
SourceAddress : 192.168.56.1
PingSucceeded : True
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded : False
```

```
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> Test-NetConnection 192.168.56.102 -Port 9000
ADVERTENCIA: TCP connect to (192.168.56.102 : 9000) failed
```

```
ComputerName      : 192.168.56.102
RemoteAddress     : 192.168.56.102
RemotePort        : 9000
InterfaceAlias    : Ethernet 2
SourceAddress     : 192.168.56.1
PingSucceeded     : True
PingReplyDetails  (RTT) : 1 ms
TcpTestSucceeded  : False
```

```
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> Test-NetConnection 192.168.56.102 -Port 8000
ADVERTENCIA: TCP connect to (192.168.56.102 : 8000) failed
```

```
ComputerName      : 192.168.56.102
RemoteAddress     : 192.168.56.102
RemotePort        : 8000
InterfaceAlias    : Ethernet 2
SourceAddress     : 192.168.56.1
PingSucceeded     : True
PingReplyDetails (RTT) : 1 ms
TcpTestSucceeded  : False
```

```
Administrador: Windows PowerShell
PS C:\WINDOWS\system32> $ip="192.168.56.102"
PS C:\WINDOWS\system32> 7000 8000 9000 | ForEach-Object {
>> $c = New-Object System.Net.Sockets.TcpClient
>> $iar = $c.BeginConnect($ip, $_, $null, $null)
>> Start-Sleep -Milliseconds 300
>> $c.Close()
>> }
PS C:\WINDOWS\system32> $ip="192.168.56.102"
PS C:\WINDOWS\system32> 7000 8000 9000 | ForEach-Object {
>> $c = New-Object System.Net.Sockets.TcpClient
>> $iar = $c.BeginConnect($ip, $_, $null, $null)
>> Start-Sleep -Milliseconds 300
>> $c.Close()
>> }
PS C:\WINDOWS\system32>

Linux Mint donar [Comando] - Oracle VM VirtualBox
Archivo Máquina Ver Entrada Dispositivos Ayuda

osboxes@osboxes:~$ sudo systemctl restart knockd
osboxes@osboxes:~$ sudo systemctl enable knockd
Synchronizing state of knockd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable knockd
osboxes@osboxes:~$ sudo systemctl restart knockd
osboxes@osboxes:~$ sudo systemctl enable knockd
Synchronizing state of knockd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable knockd
osboxes@osboxes:~$ systemctl status knockd
● knockd.service - Port-Knock Daemon
   Loaded: loaded (/usr/lib/systemd/system/knockd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-05-12 17:31:12 EDT; 16s ago
     Docs: man:knockd(1)
   Main PID: 5611 (knockd)
      Tasks: 1 (limit: 4604)
    Memory: 664.0K (peak: 726.0K)
       CPU: 13ms
    CGroup: /system.slice/knockd.service
           └─311 /usr/sbin/knockd -i mp88p

May 12 17:31:12 osboxes systemd[1]: Started knockd.service - Port-Knock Daemon.
May 12 17:31:12 osboxes knockd[5611]: starting up, listening on mp88p
osboxes@osboxes:~$ sudo journalctl -u knockd -f
May 12 17:31:12 osboxes systemd[1]: Started knockd.service - Port-Knock Daemon.
May 12 17:31:12 osboxes knockd[5611]: starting up, listening on mp88p
May 12 17:34:26 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 1
May 12 17:34:27 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 1
May 12 17:34:28 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 1
May 12 17:34:47 osboxes knockd[5611]: 192.168.56.1: closeSSH: Stage 1
May 12 17:34:48 osboxes knockd[5611]: 192.168.56.1: closeSSH: Stage 1
May 12 17:34:49 osboxes knockd[5611]: 192.168.56.1: closeSSH: Stage 1
May 12 17:35:05 osboxes knockd[5611]: 192.168.56.1: closeSSH: sequence timeout (stage 1)
May 12 17:37:52 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 1
May 12 17:37:52 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 2
May 12 17:37:52 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 3
May 12 17:37:52 osboxes knockd[5611]: 192.168.56.1: openSSH: OPEN SESAME
May 12 17:37:52 osboxes knockd[5917]: openSSH: running command: /sbin/iptables -I INPUT -s 192.168.56.1 -p tcp --dpor
t 22 -j ACCEPT
May 12 17:37:59 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 1
May 12 17:37:59 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 2
May 12 17:38:00 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 3
May 12 17:38:00 osboxes knockd[5611]: 192.168.56.1: openSSH: OPEN SESAME
May 12 17:38:00 osboxes knockd[5922]: openSSH: running command: /sbin/iptables -I INPUT -s 192.168.56.1 -p tcp --dpor
t 22 -j ACCEPT
```

```
osboxes@osboxes:~$
PS C:\WINDOWS\system32> $ip="192.168.56.102"
PS C:\WINDOWS\system32> 7000 8000 9000 | ForEach-Object {
>> $c = New-Object System.Net.Sockets.TcpClient
>> $iar = $c.BeginConnect($ip, $_, $null, $null)
>> Start-Sleep -Milliseconds 300
>> $c.Close()
>> }
PS C:\WINDOWS\system32> $ip="192.168.56.102"
PS C:\WINDOWS\system32> 7000 8000 9000 | ForEach-Object {
>> $c = New-Object System.Net.Sockets.TcpClient
>> $iar = $c.BeginConnect($ip, $_, $null, $null)
>> Start-Sleep -Milliseconds 300
>> $c.Close()
>> }
PS C:\WINDOWS\system32> ssh osboxes@192.168.56.102
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.
ED25519 key fingerprint is SHA256:qYxIkQ4T4u6vu9uPe2igJdSKp4d9gwaqPnZTFEt1JU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.102' (ED25519) to the list of known hosts.
osboxes@192.168.56.102's password:
osboxes@osboxes:~$ whoami
osboxes
osboxes@osboxes:~$ ls
Desktop Documents Downloads Music Pictures Public Servidor_Datos Templates Videos
osboxes@osboxes:~$ who
osboxes tty7      2026-05-12 17:19 (:0)
osboxes pts/1     2026-05-12 17:37
osboxes pts/2     2026-05-12 17:38 (192.168.56.1)
osboxes@osboxes:~$ ls -lh
total 36K
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Desktop
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Documents
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Downloads
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Music
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Pictures
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Public
drwxr-xr-x 4 osboxes osboxes 4.0K May 12 17:27 Servidor_Datos
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Templates
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Videos
osboxes@osboxes:~$
```

```
osboxes@osboxes:~$ ps C:\WINDOWS\system32 $ip=192.168.56.102
PS C:\WINDOWS\system32 7000 8000 3000 | ForEach-Object {
    $c = New-Object System.Net.Sockets.TcpClient
    $s = $c.BeginConnect($ip, $c, $null, $null)
    Start-Sleep -Milliseconds 300
    $c.Close()
}
PS C:\WINDOWS\system32 $ip=192.168.56.102
PS C:\WINDOWS\system32 7000 8000 3000 | ForEach-Object {
    $c = New-Object System.Net.Sockets.TcpClient
    $s = $c.BeginConnect($ip, $c, $null, $null)
    Start-Sleep -Milliseconds 300
    $c.Close()
}
PS C:\WINDOWS\system32 ssh osboxes@192.168.56.102
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.
ED25519 key fingerprint is SHA256:qYk1Q4t4u0vdupe2lg3dSKp4u0puaqPNz7FET1JU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.102' (ED25519) to the list of known hosts.
osboxes@192.168.56.102's password:
osboxes@osboxes:~$ whoami
osboxes
osboxes@osboxes:~$ ls
Desktop Documents Downloads Music Pictures Servidor_Datos Templates Videos
osboxes@osboxes:~$ who
osboxes tty7      2026-05-12 17:19 (:0)
osboxes pts/1     2026-05-12 17:37
osboxes pts/2     2026-05-12 17:38 (192.168.56.1)
osboxes@osboxes:~$ ls -lh
total 36K
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Desktop
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Documents
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Downloads
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Music
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Pictures
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Public
drwxr-xr-x 4 osboxes osboxes 4.0K May 12 17:27 Servidor_Datos
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Templates
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Videos
osboxes@osboxes:~$
```

```
Administrador: Windows PowerShell
osboxes@osboxes:~$ who
osboxes tty7      2026-05-12 17:19 (:0)
osboxes pts/1     2026-05-12 17:37
osboxes pts/2     2026-05-12 17:38 (192.168.56.1)
osboxes@osboxes:~$ ls -lh
total 36K
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Desktop
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Documents
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Downloads
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Music
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Pictures
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Public
drwxr-xr-x 4 osboxes osboxes 4.0K May 12 17:27 Servidor_Datos
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Templates
drwxr-xr-x 2 osboxes osboxes 4.0K Feb 14 2025 Videos
osboxes@osboxes:~$ exit
logout
Connection to 192.168.56.102 closed.
PS C:\WINDOWS\system32 ftp 192.168.56.2
PS C:\WINDOWS\system32 ^C
PS C:\WINDOWS\system32 ftp 192.168.56.102
Conectado a 192.168.56.102.
220 (vsFTPd 3.0.5)
200 Always in UTF8 mode.
Usuario (192.168.56.102:(none)): osboxes
331 Please specify the password.
Contraseña:
230 Login successful.
ftp> ls
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
Desktop
Documents
Downloads
Music
Pictures
Public
Servidor_Datos
Templates
Videos
226 Directory send OK.
ftp> 94 bytes recibidos en 0.00segundos 31.33a KB/s.
ftp>
```

```
Linux Mint donar [Comando] - OracleVirtualBox
osboxes@osboxes:~$ sudo systemctl restart knockd
sudo systemctl enable knockd
Synchronizing state of knockd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable knockd
Created symlink /etc/systemd/system/multi-user.target.wants/knockd.service → /usr/lib/systemd/system/knockd.service.
osboxes@osboxes:~$ sudo systemctl restart knockd
osboxes@osboxes:~$ sudo systemctl enable knockd
Synchronizing state of knockd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable knockd
osboxes@osboxes:~$ systemctl status knockd
● knockd.service - Port-Knock Daemon
   Loaded: loaded (/usr/lib/systemd/system/knockd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-05-12 17:31:12 EDT; 16s ago
     Docs: man:knockd(1)
   Main PID: 5611 (knockd)
      Tasks: 1 (limit: 4604)
     Memory: 604.0K (peak: 756.0K)
        CPU: 13ms
     CGroup: /system.slice/knockd.service
            └─c11 /usr/sbin/knockd -i mp0109

May 12 17:31:12 osboxes systemd[1]: Started knockd.service - Port-Knock Daemon.
May 12 17:31:12 osboxes knockd[5611]: starting up, listening on mp0109
osboxes@osboxes:~$ sudo journalctl -u knockd -f
May 12 17:31:12 osboxes systemd[1]: Started knockd.service - Port-Knock Daemon.
May 12 17:31:12 osboxes knockd[5611]: starting up, listening on mp0109
May 12 17:34:26 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 1
May 12 17:34:27 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 1
May 12 17:34:28 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 1
May 12 17:34:47 osboxes knockd[5611]: 192.168.56.1: openSSH: sequence timeout (stage 1)
May 12 17:34:47 osboxes knockd[5611]: 192.168.56.1: closeSSH: Stage 1
May 12 17:34:48 osboxes knockd[5611]: 192.168.56.1: closeSSH: Stage 1
May 12 17:34:49 osboxes knockd[5611]: 192.168.56.1: closeSSH: Stage 1
May 12 17:35:05 osboxes knockd[5611]: 192.168.56.1: openSSH: sequence timeout (stage 1)
May 12 17:37:52 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 1
May 12 17:37:52 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 2
May 12 17:37:52 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 3
May 12 17:37:52 osboxes knockd[5611]: 192.168.56.1: openSSH: OPEN SESAME
May 12 17:37:52 osboxes knockd[5917]: openSSH: running command: /sbin/iptables -I INPUT -s 192.168.56.1 -p tcp --dpor
t 22 -j ACCEPT
May 12 17:37:59 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 1
May 12 17:37:59 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 2
May 12 17:38:00 osboxes knockd[5611]: 192.168.56.1: openSSH: Stage 3
May 12 17:38:00 osboxes knockd[5611]: 192.168.56.1: openSSH: OPEN SESAME
May 12 17:38:00 osboxes knockd[5922]: openSSH: running command: /sbin/iptables -I INPUT -s 192.168.56.1 -p tcp --dpor
t 22 -j ACCEPT
May 12 17:38:00 osboxes knockd[5922]: openSSH: running command: /sbin/iptables -I INPUT -s 192.168.56.1 -p tcp --dpor
t 22 -j ACCEPT
osboxes@osboxes:~$ sudo nano /etc/vsftpd.conf
osboxes@osboxes:~$ sudo restart vsftpd
osboxes@osboxes:~$ sudo systemctl restart vsftpd.service
osboxes@osboxes:~$ sudo systemctl restart vsftpd
osboxes@osboxes:~$ sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable vsftpd
osboxes@osboxes:~$ sudo ss -tulpn | grep :21
tcp LISTEN 0      32             *:*          *:*          users:((('vsftpd',pid=6003,fd=3))
osboxes@osboxes:~$
```